

LETTER OF INFORMATION / CONSENT

Neural and cognitive basis of speech comprehension: MRI

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Study location	Imaging Research Centre St. Joseph's Healthcare, Fontbonne Building 301 James Street South, Hamilton, ON
Sponsor	National Institute of Deafness and Communication Disorders (United States)
Number of Participants	Approximately 250

You are invited to participate in a research study. To help you decide whether you want to participate, this form gives you detailed information about the research study. Before we start, you will be asked to sign this form. Please take your time to make your decision, ask us any questions you may have, and feel free to discuss it with your friends and family.

What are we trying to discover?

As we age, understanding speech often becomes more challenging, especially when there are background noises, for example in a busy restaurant. This study aims to learn more about how the brain perceives speech, how this changes as we get older, and why some people are more successful at comprehending speech in noise than others (even though they have similar hearing ability).

What will happen during the study?

Magnetic resonance imaging (MRI) is a technology that uses strong magnetic fields and radio frequency fields to produce detailed pictures of soft tissues in the body, including the brain. For this study, we will be using MRI to take pictures of your brain.

The main part of the experiment will use functional MRI, which measures your brain's activity. For those scans, we will ask you to listen to a story. We will occasionally pause the story to ask you questions about what you just heard. You may ask to take a short break at those times if you like. Sometimes the story will be the only sound you hear, whereas other times there will be background noises. Sometimes the voice reading the audiobook may sound muffled. You may also be asked to make judgments about the nature of what you experience.

These scans will take approximately an hour total. MRI will allow us to measure which brain regions you are using to process different aspects of speech.

Because MRI uses strong magnetic fields, we need to make sure you do not have certain metal objects in your body or with you when you enter the MRI room. You will be asked to change into a hospital gown when you arrive at the MRI facility. Your clothes and all personal items (e.g., watches, jewelry, wallet, cell phone) will be stored in a secure locker. The MR technologist will talk with you before the scanning session to answer any questions, and to make sure it is safe for you to go into the MRI.

The MRI machine looks like a big doughnut, and you will lie down on a bed with your head in the tunnel made by the “doughnut hole”. We will ensure that you are comfortable so that you can stay very still while we scan your brain to get the pictures. You should try to remain as still as possible during the scans. Movements will not be dangerous to you in any way but will blur the MRI pictures. For each MRI session, you will need to hold still in the machine for up to 6 minutes, but we can give you some small breaks to relax if you need them. The MR technologist will be able to observe you at all times. You will be able to contact the MR technologist at any time during the scan session for any reason.

You will hear moderately loud knocking or beeping sounds when the MRI machine is scanning. You will be given ear protection to wear in the MRI. Different types of scans will make different types of sounds, which is normal for MRI. The technologist will talk to you before each scan starts. There will be a mixture of short scans and some longer scans (up to 6 minutes each).

Are there any risks to doing this study?

Metal Objects. Before you can participate in an MRI study, we need to make sure it is safe for you to do so. Because certain metal objects may lead to injuries during the MRI procedure, we will ask you to answer questions about any metal implants or objects you might have in your body and the location of any tattoos (which may contain ink that would interact with magnetic fields). If you have any metal implants or objects that are not safe for the 3T MRI at St. Joseph's Healthcare Hamilton, you will not be allowed to be scanned. Some objects that are not safe for MRI include: cardiac pacemakers, metal fragments in the eye, aneurysm clips in your brain. The research study staff and the MR technologist will work together to make sure you will be safe in the scanner. We will also ask whether you are extremely uncomfortable in enclosed spaces (claustrophobia).

Long-term risks. Based on the use of MRI in medicine for over 20 years, imaging experts believe there are no long-term negative health effects caused by the magnetic field strength used in this study. This MRI study does not involve any form of harmful ionizing radiation or injections.

Other risks. Some people may feel uncomfortable lying still in the confined space of the MRI scanner, tingling sensations are felt by some people during certain scans, or you may feel dizzy for a few minutes at the end of the MRI study. These are infrequent but expected sensations.

It is important you understand that you will be able to contact the technologist at any time during the scan. You may ask to be taken out of the scanner for any reason without any penalty or consequences.

There is a small risk of a breach of privacy. If such a breach should happen, others would be able to link your identity to the data we collect. I describe below the steps I am taking to protect your privacy (see *Confidentiality*).

Unexpected findings. In the unlikely event that an abnormal finding is seen on your MRI scan, we will ask a radiologist to look at your scan. It is important to realize that research scans are not designed to be used for diagnosis, and no formal report will be issued. Your identity will not be revealed to the radiologist. If the radiologist recommends further tests to determine the nature and significance of any incidental findings on your MRI scan, we will relay this information to you and advise you to follow up with your family doctor.

Pregnancy. Pregnant women are not candidates for research MRI studies. As with medications and other imaging procedures, it is considered prudent not to undergo MRI during pregnancy unless there is a medical need. If you are a woman of child-bearing age, we will ask you to confirm that you are not pregnant, nor likely to be pregnant, at the time of the study.

Are there any benefits to doing this study?

The research will not benefit you directly, but will help scientists understand how people comprehend speech. In the long term, this may help us develop better treatments or devices that help people communicate in noisy environments.

Payment or Reimbursement

You will be reimbursed \$30 per hour for your participation in the experiment. We anticipate the study to last approximately 2 hours. If you come by car we will provide you with a parking pass, otherwise we will reimburse you \$10 for travel.

Confidentiality

To protect your privacy, your name, date of birth and other directly identifying information will be removed from the dataset and stored separately and securely in Dr. Brodbeck's lab. Your responses and data will be logged under an anonymous code. *Data that we believe may allow others to determine who you are, such as your facial features, will be irreversibly changed through a family of processes called anonymization so that you cannot be identified.* No information that discloses your identity will be released without your consent, unless required by law (for example, laws pertaining to the protection of vulnerable individuals including children).

What if I change my mind about being in the study?

Your participation in this study is voluntary. If you do not want to answer some of the questions you do not have to, but you can still be in the study. You can decide to stop at any time, even after signing the consent form or part-way through the study. If you decide to stop, there will be no consequences to you, and you have the option of removing your data from the study.

How do I find out what was learned in this study?

If you would like a summary of the study results, please let me know how you would like it sent to you.

Questions about the Study

If you have questions or need more information about the study itself, please contact us at brodbecc@mcmaster.ca

This study has been reviewed by the Hamilton Integrated Research Ethics Board (HiREB). The HiREB is responsible for ensuring that participants are informed of the risks associated with the research, and that participants are free to decide if participation is right for them. If you have any questions about your rights as a research participant, please call the Office of the Chair, HiREB, at 905.521.2100 x 42013.

For the purposes of ensuring proper monitoring of the research study, it is possible that representatives of HiREB, this institution, and affiliated sites or regulatory authorities may consult your original (identifiable) research data to check that the information collected for the study is correct and follows proper laws and guidelines. By participating in this study, you authorize such access.

CONSENT

I acknowledge that the research procedures have been explained to me and that any questions that I asked have been answered to my satisfaction. I have been informed of the alternatives to participation in this study, including the right not to participate and the right to withdraw at any time without prejudice. The potential harms and benefits of participating have been explained to me. I know that I may ask now, or in the future, any questions I have about the study or the research procedures. I have been assured that individual results will be kept confidential, and that no information will be released or printed that would disclose my personal identity without my permission, unless required by law. I understand that I will be offered a copy of this form to keep.

I agree to participate in the study:

_____	_____	_____
Name of Participant (Printed)	Signature	Date

Consent form explained in person by:

_____	_____	_____
Name and Role (Printed)	Signature	Date

CONSENT: Future use and data sharing

Future use

With your permission, the data from this study may be used for other, future research projects by Dr. Brodbeck. Those future projects may focus on topics that are different from this study, but they will likely be related to improving our understanding of how the human brain works.

☐ I agree that my data may be used for future research by Dr. Brodbeck.

Sharing of de-identified data

With your permission, we will share this dataset with other researchers by uploading it to a public online repository after we complete the study.

After having gone through the described de-identification process, we will upload a dataset made of your de-identified data and those of other participants to OpenNeuro. OpenNeuro is a database developed at Stanford University and funded by the National Institutes of Health as part of the BRAIN Initiative. OpenNeuro aims to make scientific data as easily accessible as possible to the scientific community while respecting your privacy. OpenNeuro is publicly available to anyone with internet access. Researchers from around the world will be able to use the de-identified data from this research study and other studies. This maximizes the potential for this data to be used for additional research that may lead to important scientific discoveries.

By agreeing to participate, you will be making a free and generous gift for research that might help others. It is possible that some of the research conducted using your information eventually could lead to the development of new methods for studying the brain, new diagnostic tests, new drugs, or other commercial products. Should this occur, there is no plan to provide you with any part of the profits generated from such products and you will not have any ownership rights in the products.

Your privacy will be protected through advanced de-identification techniques. Despite this, there is always a small risk that your shared data may one day enable someone to identify you. For example, someone could use information in your clinical record to match you in a dataset. This is difficult to do because of the de-identification measures. Extensive sharing of personal information through social media or genealogical websites may also make it easier to re-identify you. In the remote chance that you are re-identified, you may suffer a loss of privacy or potentially be subject to discrimination.

☐ I agree that my anonymized data may be included in a dataset released to the public.

Name of Participant (Printed)

Signature

Date

Consent form explained in person by:

Name and Role (Printed)

Signature

Date